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Demonstration of low-overhead quantum error correction codes

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Quantum computers hold the potential to surpass classical computers in solving complex computational problems. The fragility of quantum information and the error-prone nature of quantum operations necessitate the use of quantum error correction codes to achieve fault-tolerant quantum computing. However, most codes that have been demonstrated so far suffer from low encoding efficiency, and their scalability is hindered by prohibitively high resource overheads. Here we use a 32-qubit quantum processor to demonstrate two low-overhead quantum low-density parity-check codes, a distance-4 bivariate bicycle code and a distance-3 punctured bivariate bicycle code. Utilizing a two-dimensional architecture with overlapping long-range couplers connecting the qubits, we demonstrate the simultaneous measurements of all non-local weight-6 stabilizers via the periodic execution of an efficient syndrome extraction circuit. We achieve a logical error rate per logical qubit per cycle of (8.91 ± 0.17)% for the bivariate bicycle code with four logical qubits and (7.77 ± 0.12)% for the punctured bivariate bicycle code with six logical qubits. Our results establish the feasibility of performing quantum error correction with long-range coupled superconducting processors, demonstrating the viability of low-overhead quantum error correction.

Quantum computing promises computational advantages in a wide range of applications[1], including factoring[2], optimization[3], chemistry[4], finance[5] and machine learning[6,7]. Yet, for practical quantum computation to attain such advantages, errors on the physical level of the device need to be corrected so that large and deep circuits can be executed with the desired accuracy. Quantum error correction provides a way to reach quantum fault tolerance through encoding logical qubits redundantly into many physical qubits, which enables one to diagnose and correct errors by repeatedly measuring error syndromes[8,9]. Experimental demonstrations of different quantum error correction codes have been reported in diverse platforms, such as trapped ions[10–14], nitrogen-vacancy centres[15–17], neutral atoms[18,19] and superconducting processors[20–30].

A crucial aspect of quantum error correction codes concerns their encoding efficiency. Many existing codes fall short in this aspect, including the surface code[9,31,32] that for 20 years was the leading

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approach towards fault-tolerant quantum computing. Indeed, in the surface code, a single logical qubit is encoded into a $d \times d$ square of physical qubits, leading to an encoding rate scaling as $1/d^2$ and vanishing in the large-d limit. This compromises its scalability to hundreds or more logical qubits, despite the fact that small-scale surface codes with a single logical qubit have been demonstrated by several groups[20–23,28–30]. An intriguing way out of this dilemma is to consider more general quantum low-density parity-check (qLDPC) codes[33], which would achieve quantum fault tolerance with a much higher encoding efficiency. In particular, a special family of qLDPC codes, known as bivariate bicycle (BB) codes, has recently been proposed and shown to exhibit high encoding efficiency[34]. Under the circuit-based noise model, these codes merit an error threshold close to 0.7%, which is comparable with that of the surface code. In addition, they support a low-depth syndrome measurement circuit, an efficient decoding algorithm and a fault-tolerant protocol for implementing logical gates. Despite these merits and the fact that hardware requirements for realizing BB codes are relatively mild compared with that for several other qLDPC codes such as hypergraph product[35] or quantum Tanner codes[36,37], the implementation of BB codes is exceedingly more challenging than that of the surface code and, thus, has evaded experiment so far. In fact, BB codes pose a more stringent demand on designing a category of superconducting processors with higher connectivity and more freedom in control, greatly surpassing the conventional two-dimensional architecture[30,38,39] in which remotely separated physical qubits must possess precisely programmable couplings to allow for direct gate operations.

Here we overcome these difficulties and report the demonstration of a distance-4 BB code and a distance-3 punctured BB code on our latest superconducting processor, Kunlun, with 32 long-range-coupled transmon qubits (Fig. 1). We design and fabricate our Kunlun processor to bear a torus connectivity topology, which is adapted to planar architectures by integrating multilength tunable couplers for distant qubits separated by length scales up to 6.5 mm (Fig. 1a). Our device supports programmable interconnects among qubits with a maximum degree of six, meeting the connectivity requirement for implementing these codes. The average fidelities for the parallel single- and two-qubit gates are 99.95% and 99.22%, respectively. We repeatedly execute an efficient syndrome measurement circuit over multiple cycles to extract error syndromes, which are then decoded using the belief propagation with an ordered statistics decoder (BP-OSD)[40,41] extended to the circuit-based noise model. Our work paves the way for implementing diverse low-overhead qLDPC codes with superconducting processors.

The BB code

BB codes are a family of Calderbank–Shor–Steane quantum stabilizer codes constructed from bivariate polynomials[34]. We use the notation $[[n, k, d]]$ to describe an error correction code that encodes k distance–d logical qubits into the joint entangled state of n data qubits. For our experimental demonstration, we have designed a BB code with parameters [[18, 4, 4]]. The four logical qubits are defined by a set of logical Pauli operators $\{\bar{X}_1, \bar{Z}_1 \ldots \bar{X}_4, \bar{Z}_4\}$. This distance-4 BB code can be described by a Tanner graph embedded on a torus with periodic boundary conditions (Fig. 1a), where vertices correspond either to data (for encoding logical qubits) or check (for measuring error syndromes) qubits. An edge connecting a check qubit and a data qubit indicates that the corresponding check operator acts non-trivially on the data qubit. Each check operator is geometrically non-local and acts non-trivially on six data qubits—four local and two long range. All X-type (or Z-type) check operators can be obtained through horizontal and vertical translations of a single one across the grid.

By partitioning the check matrices into the left and right parts (Methods), the 18 data qubits can be accordingly categorized into L type and R type. Each data qubit is labelled either as L^i or R^i, where i ranges from 1 to 9. The check qubits are labelled as X^j (measuring X-type stabilizers) or Z^j (measuring Z-type stabilizers), with j ranging from 1 to 9. We note that the direct construction of BB codes includes a total of 18 check operators, but only 14 of them are independent. We, thus, remove four redundant check operators and retain only the independent ones to reduce experimental complexities. The punctured BB code with parameters [[18, 6, 3]] is constructed by further removing two check operators from the 14 independent check operators of the BB code (Methods).

Experimental setup and framework

Our experiments are performed on a flip-chip superconducting quantum processor with 32 frequency-tunable transmon qubits arranged in a square lattice (Fig. 1a). The average lifetime of all physical qubits is measured to be $T_1 = 41.8$ μs. To implement the distance-4 BB code, we use 18 qubits as data qubits and the remaining 14 as check qubits. The distance-3 punctured BB code is implemented using 18 data qubits and 12 check qubits. Compared with the BB code, the check qubits X^5 and Z^5 and the couplers connecting them to the data qubits are not used (Methods). The processor possesses 84 tunable couplers with lengths ranging from 1 mm to 6.5 mm, offering flexible interconnects between the check and data qubits with lattice distances up to five. In particular, each check qubit is coupled to six data qubits via both local and long-range couplers. To realize the torus connectivity topology of the degree-6 Tanner graph on a planar architecture, we introduce up to 15 air bridges[42,43] for each coupler to enable its crossovers with the other couplers (Supplementary Section II.A).

In our experiment, the stabilizer measurement circuits are compiled using Hadamard and CZ gates, with additional Pauli X and Y gates inserted to reduce the dephasing errors. Single-qubit gates are realized by applying resonant microwave pulses with a duration of 30 ns and have an average fidelity of 99.95%, as determined by simultaneous cross-entropy benchmarking among all qubits. The 84 pairs of CZ gates have a duration of 105 ns each, including two 5-ns-long idling times at the beginning and the end, and exhibit an average fidelity of 99.22% (Fig. 1b). To execute the syndrome measurement circuit for multiple cycles, it is crucial to reduce the readout times of the check qubits, as the data qubits can suffer from decoherence when measuring the check qubits. We realize a readout duration of 920 ns for the check qubits, including a 520-ns-long readout pulse and another 400-ns-long idling time for photon decaying (Fig. 1c). As a result, we obtain an average readout error of 1.6% for simultaneous two-computational-state readout and 6.6% for simultaneous three-state readout, which discriminates between the two computational states and a leakage state.

We integrate all individual stabilizer measurements into a syndrome measurement circuit, enabling the simultaneous extraction of all stabilizer values. As illustrated in Fig. 1d and Extended Data Fig. 1, the full cycle of the syndrome measurement circuit includes qubit-state initialization, seven layers of non-overlapping CZ gates, eight layers of single-qubit gates and qubit readout. We characterize the fidelity of seven layers of non-overlapping CZ gates using simultaneous cross-entropy benchmarking, with the cumulative distributions of error probabilities shown in Fig. 2a. We observe a similar performance of CZ gates among qubit pairs interconnected by either local or long-range couplers. To protect qubits from dephasing, we insert single-qubit gates between CZ gates and apply dynamical decoupling (DD) gates during qubit idles (Supplementary Section II.D).

With all elements in place, we implement the distance-4 BB code for logical-state preservation. In a single experimental run, we begin with encoding a four-logical-qubit state. To encode a logical Z basis state, we first initialize each data qubit randomly into either $|0\rangle$ or $|1\rangle$. Subsequently, we execute one cycle of stabilizer measurements, setting the four-logical-qubit state into $|\phi_1\phi_2\phi_3\phi_4\rangle_{\bar{L}}$. Each ϕ_i is determined by the parity of the corresponding logical operator $\bar{Z}_i$: $\phi_i = 0$ for even parity and $\phi_i = 1$ for odd parity. Preparing logical X basis states follows a similar procedure. After encoding, we execute multiple consecutive syndrome

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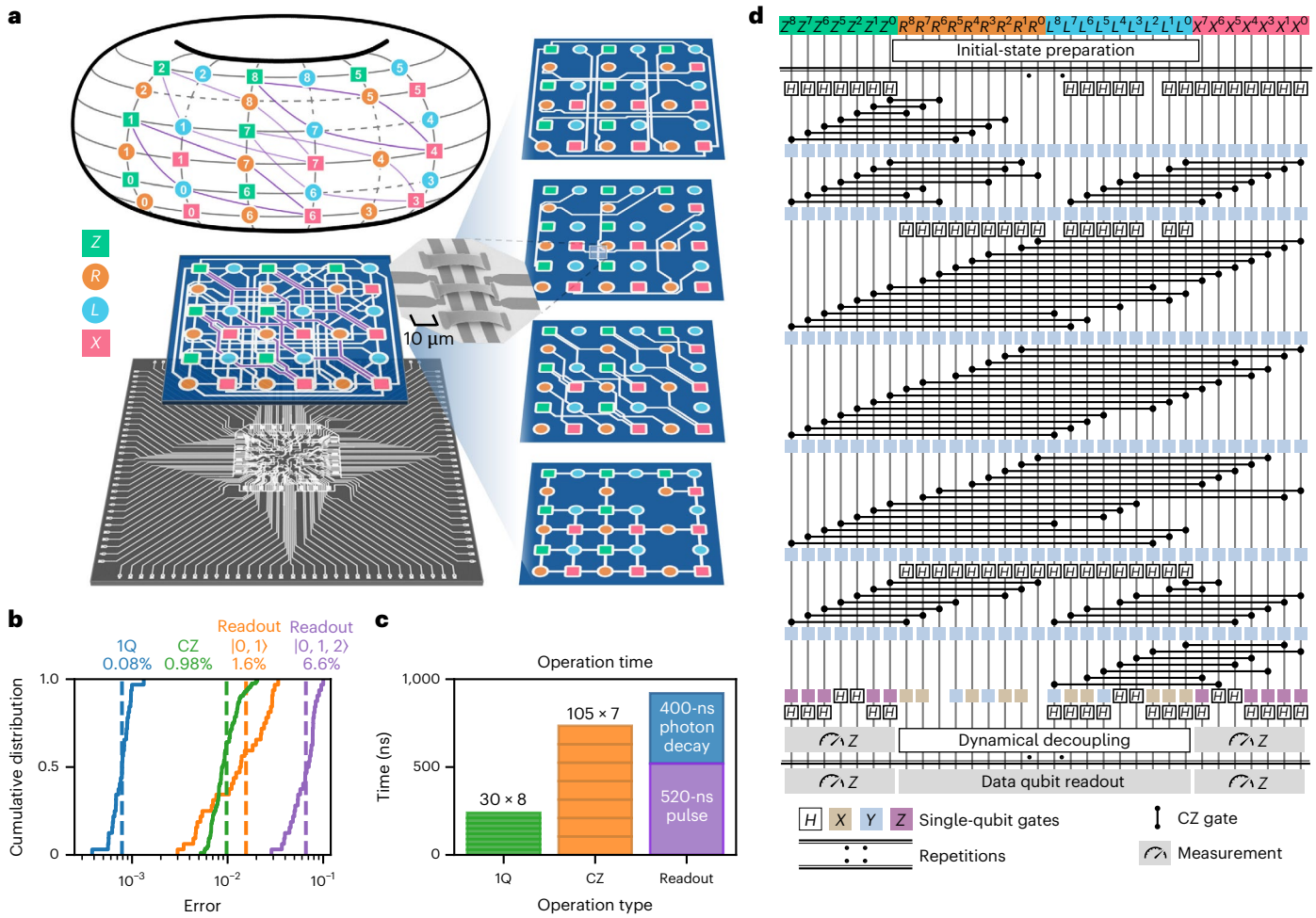


Fig. 1 | Implementation of the BB code. a, Schematic of the Kunlun processor. Top left: Tanner graph of the BB code embedded into a torus, where relevant sites are indexed by colour-coded shapes and numbers. Only part of long-range connections are shown. Bottom left: corresponding layout of the 32-qubit superconducting quantum processor Kunlun that implements the [[18, 4, 4]] BB code. The processor comprises 18 data qubits (circles) categorized into *L* type (blue) and *R* type (orange), and 14 check qubits (squares) for the *X*-type (red) and *Z*-type (green) stabilizer measurements. The purple lines highlight couplers corresponding to the long-range connections in the Tanner graph. Right: 84 couplers are grouped into four sets according to their lengths, and a crossing region between two interleaved couplers is magnified to reveal the underlying microstructure. **b**, Cumulative distributions of error probabilities measured on Kunlun. Blue, Pauli errors for parallel single-qubit gates; green, Pauli errors for parallel CZ gates; orange (purple), assignment errors for simultaneous two-state (three-state) measurements. Dashed vertical lines denote the mean values. **c**, Operation times of quantum operations in a single error correction cycle. The total cycle duration is 1,895 ns, including eight 30-ns single-qubit gate layers, seven 105-ns CZ gate layers, a 520-ns readout pulse and a 400-ns photon ring-down time. The measurement of the data qubit in the last cycle is performed with an 890-ns readout pulse. **d**, Syndrome measurement circuit of the [[18, 4, 4]] BB code. The data qubits (labelled L^i and R^i) are initially prepared in the product states. The first syndrome cycle of the stabilizer measurements projects the data qubits into a logical state, and the subsequent cycles extract stabilizers for error correction. Each syndrome cycle contains seven layers of CZ gates interleaved with single-qubit gates to simultaneously extract *X*-type and *Z*-type stabilizers, and ends with the measurement of the *X*-type and *Z*-type check qubits (labelled X^i and Z^i). In the final cycle, the data qubits are measured to obtain both stabilizer values and information of the logical state. We insert additional Pauli X and Y gates between CZ layers and implement DD during the measurement of the check qubits to protect the data qubits from dephasing.

cycles, recording the stabilizer outcomes to construct the error syndromes (Methods). In the final cycle, we directly measure all data qubits to extract both stabilizer parity information and logical measurement outcomes. We decode the error syndromes to infer the most likely error configuration and then apply corrections in post-processing. An experimental instance is considered successful if the corrected logical state agrees with the initial encoded logical state; otherwise, a logical error is recorded. To evaluate the logical performance, we perform up to six consecutive syndrome cycles. For each cycle $t = 1 \ldots 6$, we collect data from over 40,000 experimental instances (excluding instances in which leakage is detected) and decode the error syndromes to obtain the logical error probability. We fit the logical error probability as a function of the number of executed syndrome cycles to extract the logical error per cycle.

Error detection

Physical errors in the syndrome measurement circuit can be detected by comparing stabilizer values across consecutive syndrome cycles, based on which the error syndromes are constructed. Any difference between a stabilizer's value and its corresponding value in the preceding cycle indicates an error. To characterize the distribution of errors and implementing the distance-4 BB code, we present the detection probabilities across seven syndrome cycles, including the initial cycle for logical-state preparation. The results shown in Fig. 2b are based on data aggregated from more than 40,000 experimental instances, excluding any experimental runs in which leakage is detected. The average detection probability is 0.319 for *Z*-type stabilizers and 0.318 for *X*-type stabilizers. The lower detection probabilities observed in the initial and final cycles arise from boundary effects[44]. In the initial

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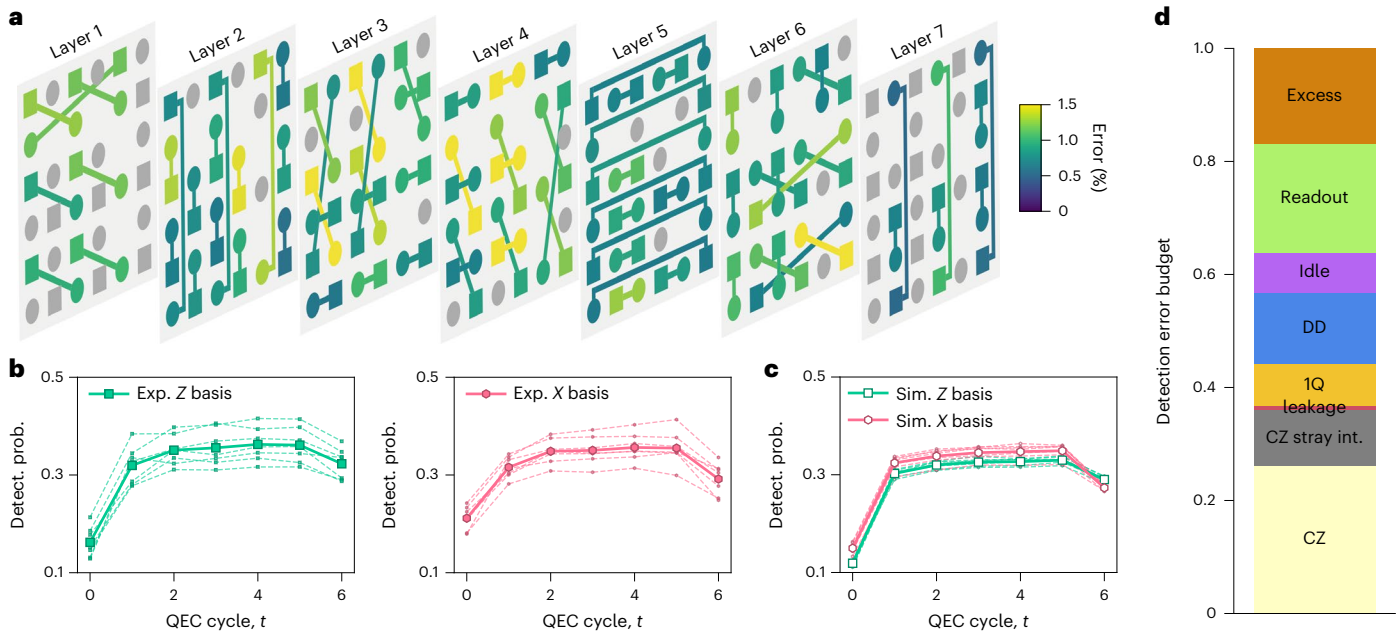


Fig. 2 | Error detection in the BB code. a, Characterized CZ errors using simultaneous cross-entropy benchmarking, where the seven CZ layers correspond to those in a syndrome measurement cycle shown in Fig. 1d. **b**, Error detection probability over seven syndrome cycles for logical *Z*-basis and *X*-basis preservation experiments (more than 40,000 experimental instances after leakage rejection). Dashed lines represent individual stabilizers and the solid line represents the average over all stabilizers. **c**, Similar to the data in **b**, but using the Pauli+ simulation. **d**, Detection error budget based on Pauli+ simulations. CZ, contributions from CZ error (excluding leakage and stray interactions); CZ stray int., unwanted interactions between parallel CZ gates; leakage, leakage during CZs and due to passive heating; 1Q, single-qubit gate; DD, decoherence during DD; idle, non-DD idle gate; excess, additional depolarizing errors appended after DDs.

cycle, stabilizer values are compared with the expected values determined by the initial-state preparation of data qubits. In the final cycle, stabilizer values are directly obtained by the data qubit readout. Both processes introduce fewer errors compared with a full syndrome cycle. The experimental results for the punctured BB code are presented in Extended Data Fig. 2. The average detection probability is 0.312 for both *X*-type and *Z*-type stabilizers, closely matching the values observed in the BB code.

To help understand the impact of different experimental error sources, we use the Pauli+ framework[23] to simulate the BB code experiment. The simulation incorporates error sources from single-qubit gates, CZ gates, readout, idling, DD, leakage and stray interactions between parallel CZ gates (Supplementary Section I.D). As the DD sequences are not individually benchmarked in our experiment, the exact error rate of DD operations could not be determined. To account for this effect in the simulation, we introduce an additional depolarizing channel with a rate of 3.0% on each data qubit during DD (920-ns duration). This contribution, referred to as the 'excess' error, is treated separately from the decoherence error during DD. We emphasize that this excess error channel provides a phenomenological description of unmodelled processes (for example, stray interactions between neighbouring qubits during measurements), and we leave the exploration of the precise underlying physics for future studies.

The numerical results are shown in Fig. 2c. We find that the average detection probability is 0.288 for *Z*-type stabilizers and 0.304 for *X*-type stabilizers, both being close to the corresponding experimental values. The simulation for the punctured BB code yields an average detection probability of 0.284 and 0.307 for *Z*- and *X*-type stabilizers, respectively (Extended Data Fig. 2). The close agreement of detection error probabilities between the BB code and the punctured BB code (both in experiment and simulation) is expected, as both codes share similar structures, are simulated using the same parameters and are implemented on the same device. To quantify the relative contribution of different error types to the detection error of the BB code, we conduct a detection error budget analysis (Methods). The results (Fig. 2d) suggest that CZ errors and readout errors are the dominant contributors to the detection error. Compared with recent experimental realizations of surface codes[22,23], the error detection probabilities in the BB code and the punctured BB code are higher. In addition to factors such as CZ and readout errors, we attribute the higher detection probabilities to weight-6 stabilizers in the BB code. Errors detected by a weight-6 stabilizer may originate from any of the six coupled data qubits, thereby resulting in a higher detection probability than that of weight-4 or weight-2 stabilizers typically used in surface codes.

Decoding and logical error rates

To examine the logical performance of the distance-4 BB code and the distance-3 punctured BB code, we decode the error syndromes and correct the inferred errors in post-processing. For decoding, we use the BP-OSD decoder extended to the circuit-based depolarizing noise model. After decoding and applying correction, we compute the logical error probabilities $P_L(t)$ as a function of the number of executed cycles t, based on more than 40,000 experimental instances (after leakage rejection).

The results are shown in Fig. 3a,b. To extract the logical error rate per cycle p_L, we fit the data using the relation $1 - P_L \propto (1 - p_L)^t$. We then estimate the logical error rate per logical qubit per cycle as $\varepsilon_L = 1 - (1 - p_L)^{1/k}$, where k is the number of logical qubits encoded in the implemented code. For the BB code, we obtain a logical error rate per logical qubit per cycle of (9.15 ± 0.27)% and (8.67 ± 0.21)% for logical *Z*-basis and *X*-basis state preservation, respectively, with an average value of (8.91 ± 0.17)%. The error bar indicates one standard deviation for statistical and fit uncertainty. For the punctured BB code (Extended Data Fig. 3), similar fitting yields a logical error per logical qubit per cycle of (7.79 ± 0.12)% for *Z* bases and (7.75 ± 0.21)% for *X* bases, with an average of (7.77 ± 0.12)%. For reference, we use the largest measured lifetimes T_1 and T_2 among all physical qubits to estimate the physical error rate per qubit per cycle. The resulting values are

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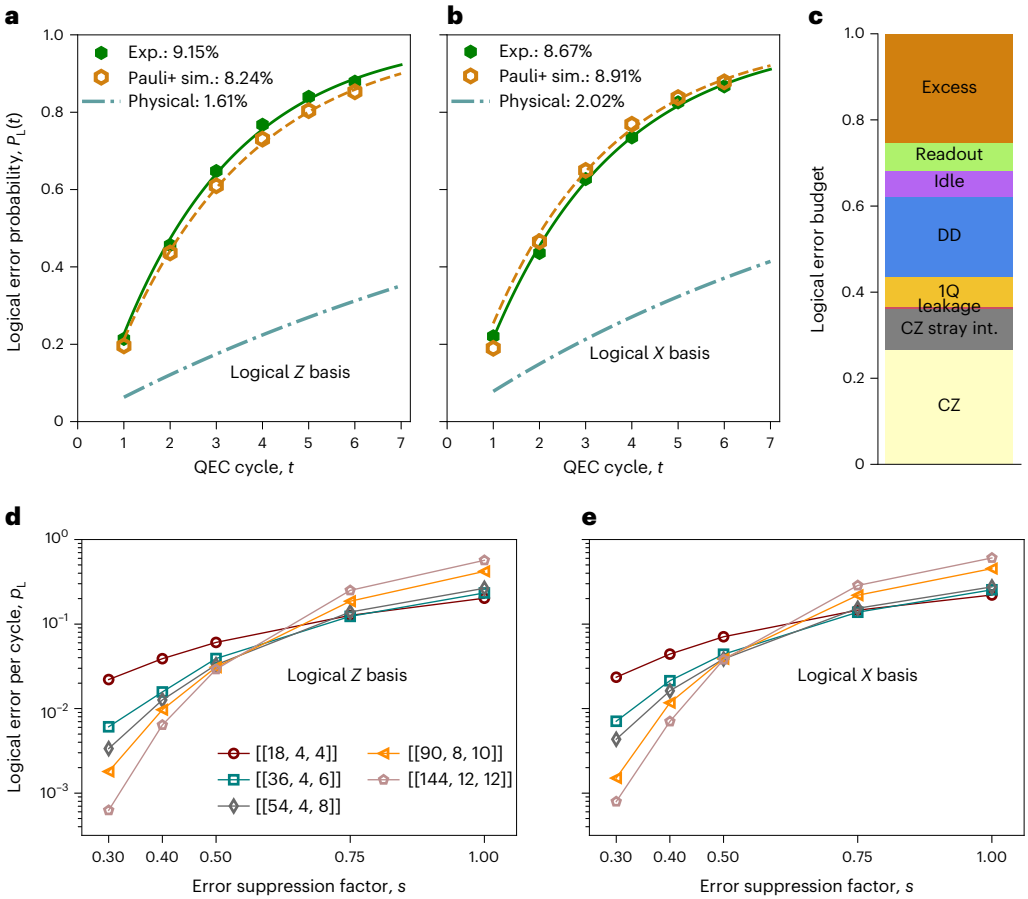


Fig. 3 | Logical error rate per cycle and performance prediction. a,**b**, Logical error probability $P_L(t)$ (filled symbols: experiment; empty symbols: Pauli+ simulation) versus the number of executed cycles, for logical Z basis state preservation (**a**) and logical X basis state preservation (**b**). Each individual data point represents more than 40,000 experimental or simulated instances. Solid line: fit to the experimental data from $t = 1$ to $t = 6$. Dashed line: fit to the simulation. The fit is performed using the least-squares method on $\log[1 - P_L(t)]$ versus t. Dash–dotted line: the estimated accumulated errors for four independent physical qubits are shown for reference. QEC, quantum error correction. **c**, Logical error budget based on Pauli+ simulations. **d**,**e**, Simulated logical error rate per cycle p_L of BB codes as functions of code distance and an error suppression factor s on the current physical error rates exhibited in Fig. 1c (semilog plot). $s = 1$ corresponds to the physical error rates in the current device.

$[1 - \exp(-t_c/T_1)]/2 = 1.61\%$ and $[1 - \exp(-t_c/T_2)]/2 = 2.02\%$ for Z- and X-basis state preservation, respectively, where t_c denotes the cycle duration. These results indicate that the break-even point is not yet achieved for either code. To quantify the relative contributions of individual error types to the logical error, we conduct a logical error budget (Methods). The results (Fig. 3c) show that CZ errors and errors during DD (including both decoherence and excess errors) are dominant contributors. We expect to suppress these dominant error sources in future devices by improving the coherence times, strengthening the qubit–coupler coupling to reduce the CZ gate time, and shortening the measurement time of the check qubit to decrease the overall DD duration.

To estimate the logical performance of BB codes in future devices with improved performance, we simulate the experiment of BB codes with distances ranging from 4 to 12 under varying physical error rates. The specific code parameters of BB codes used in the simulations are provided in Extended Data Table 1. For efficiency, only Pauli errors are included in the simulation. We introduce a suppression factor s to represent that each physical error rate used in the simulation is scaled to s times the corresponding value in the current device (Fig. 1b). The simulation results (Fig. 3d,e) indicate that at the current noise level ($s = 1$), the logical error rate increases with code distance, suggesting that codes with a larger distance do not yet benefit from logical error suppression. By contrast, when physical error rates are sufficiently reduced, increasing the code distance would lead to a suppression of logical error rates. The transition between these two regimes occurs at approximately $s = 0.5$. It is worth noting that this analysis relies on a simple circuit-based noise model. A more accurate evaluation should account for additional experimental imperfections, such as crosstalk and leakage, to better characterize real devices.

Discussion

In summary, we have experimentally demonstrated two low-overhead qLDPC codes with our Kunlun superconducting processor. In terms of encoding efficiency for use as a logical memory, implementing four distance-4 surface codes would require 124 physical qubits in total—nearly four times the number of qubits in the distance-4 BB code implementation. The success of our experiments crucially relies on high-fidelity CZ gates between distant transmon qubits achieved on the Kunlun processor by using overlapping long-range couplers. The controllability and long-range connectivity demonstrated in our experiment clearly manifest the potential of realizing other qLDPC codes with the superconducting platform, which we leave for future investigations.

Several challenges remain. First, the suppression of logical error requires that the physical error rate be below a critical threshold. In our current experiments, this has not been achieved yet, and the logical error rate remains larger than that of the physical qubits for both distance-4 BB code and distance-3 punctured BB code. To attain and

go beyond the break-even point, further technical improvements are demanded, including the realization of two-qubit gates with even higher fidelities and on qubit pairs separated by even larger distances, faster and more accurate readout of check qubits[45–49], as well as a more efficient leakage removal method[50,51]. Second, practical fault-tolerant quantum computing demands that a universal logical gate set can be implemented in a fault-tolerant fashion, which is exceedingly challenging and has so far evaded experimental demonstration for both the surface code and qLDPC codes considered in the current work. Third, it is also of crucial importance to scale up our experiments to larger low-overhead qLDPC codes and demonstrate the exponential suppression of logical errors with increasing code distance. This would mark another key step towards large-scale, fault-tolerant quantum computing.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41567-025-03157-4.

References

1. Nielsen, M. A., Chuang, I. L. *Quantum Computation and Quantum Information* (Cambridge Univ. Press, 2010).
2. Shor, P. W. Scheme for reducing decoherence in quantum computer memory. *Phys. Rev. A* **52**, R2493–R2496 (1995).
3. Farhi, E. et al. A quantum adiabatic evolution algorithm applied to random instances of an NP-complete problem. *Science* **292**, 472–475 (2001).
4. Kandala, A. et al. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature* **549**, 242–246 (2017).
5. Herman, D. et al. Quantum computing for finance. *Nat. Rev. Phys.* **5**, 450–465 (2023).
6. Biamonte, J. et al. Quantum machine learning. *Nature* **549**, 195–202 (2017).
7. Das Sarma, S., Deng, D.-L. & Duan, L.-M. Machine learning meets quantum physics. *Phys. Today* **72**, 48–54 (2019).
8. Gottesman, D. E. *Stabilizer Codes and Quantum Error Correction*. PhD thesis, California Institute of Technology (1997).
9. Dennis, E., Kitaev, A., Landahl, A. & Preskill, J. Topological quantum memory. *J. Math. Phys.* **43**, 4452–4505 (2002).
10. Flühmann, C. et al. Encoding a qubit in a trapped-ion mechanical oscillator. *Nature* **566**, 513–517 (2019).
11. Egan, L. et al. Fault-tolerant control of an error-corrected qubit. *Nature* **598**, 281–286 (2021).
12. Ryan-Anderson, C. et al. Realization of real-time fault-tolerant quantum error correction. *Phys. Rev. X* **11**, 041058 (2021).
13. de Neeve, B., Nguyen, T.-L., Behrle, T. & Home, J. P. Error correction of a logical grid state qubit by dissipative pumping. *Nat. Phys.* **18**, 296–300 (2022).
14. Paetznick, A. et al. Demonstration of logical qubits and repeated error correction with better-than-physical error rates. Preprint at https://arxiv.org/abs/2404.02280 (2024).
15. Waldherr, G. et al. Quantum error correction in a solid-state hybrid spin register. *Nature* **506**, 204–207 (2014).
16. Abobeih, M. H. et al. Fault-tolerant operation of a logical qubit in a diamond quantum processor. *Nature* **606**, 884–889 (2022).
17. Chang, X.-Y. et al. Hybrid entanglement and bit-flip error correction in a scalable quantum network node. *Nat. Phys.* **22**, 583–589 (2025).
18. Bluvstein, D. et al. Logical quantum processor based on reconfigurable atom arrays. *Nature* **626**, 58–65 (2024).
19. Reichardt, B. W. et al. Logical computation demonstrated with a neutral atom quantum processor. Preprint at https://arxiv.org/abs/2411.11822 (2024).
20. Marques, J. F. et al. Logical-qubit operations in an error-detecting surface code. *Nat. Phys.* **18**, 80–86 (2022).
21. Zhao, Y. et al. Realization of an error-correcting surface code with superconducting qubits. *Phys. Rev. Lett.* **129**, 030501 (2022).
22. Krinner, S. et al. Realizing repeated quantum error correction in a distance-three surface code. *Nature* **605**, 669–674 (2022).
23. Google Quantum AI et al. Suppressing quantum errors by scaling a surface code logical qubit. *Nature* **614**, 676–681 (2023).
24. Ni, Z. et al. Beating the break-even point with a discrete-variable-encoded logical qubit. *Nature* **616**, 56–60 (2023).
25. Sivak, V. V. et al. Real-time quantum error correction beyond break-even. *Nature* **616**, 50–55 (2023).
26. Gupta, R. S. et al. Encoding a magic state with beyond break-even fidelity. *Nature* **625**, 259–263 (2024).
27. Lacroix, N. et al. Scaling and logic in the colour code on a superconducting quantum processor. *Nature* **645**, 614–619 (2025).
28. Caune, L. et al. Demonstrating real-time and low-latency quantum error correction with superconducting qubits. Preprint at https://arxiv.org/abs/2410.05202 (2024).
29. Eickbusch, A. et al. Demonstration of dynamic surface codes. *Nat. Phys.* **21**, 1994–2001 (2025).
30. Google Quantum AI and Collaborators et al. Quantum error correction below the surface code threshold. *Nature* **638**, 920–926 (2025).
31. Kitaev, A. Y. Fault-tolerant quantum computation by anyons. *Ann. Phys.* **303**, 2–30 (2003).
32. Bravyi, S. B. & Kitaev, A. Y. Quantum codes on a lattice with boundary. Preprint at https://arxiv.org/abs/quant-ph/9811052 (1998).
33. Breuckmann, N. P. & Eberhardt, J. N. Quantum low-density parity-check codes. *PRX Quantum* **2**, 040101 (2021).
34. Bravyi, S. et al. High-threshold and low-overhead fault-tolerant quantum memory. *Nature* **627**, 778–782 (2024).
35. Tillich, J.-P. & Zémor, G. Quantum LDPC codes with positive rate and minimum distance proportional to the square root of the blocklength. *IEEE Trans. Inf. Theory* **60**, 1193–1202 (2014).
36. Panteleev, P. & Kalachev, G. Asymptotically good quantum and locally testable classical LDPC codes. In *Proc. 54th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2022)* 375–388 (ACM, 2022).
37. Leverrier, A. & Zemor, G. Quantum Tanner codes. In *Proc. 63rd Annual IEEE Symposium on Foundations of Computer Science (FOCS 2022)* 872–883 (IEEE, 2022).
38. Kim, Y. et al. Evidence for the utility of quantum computing before fault tolerance. *Nature* **618**, 500–505 (2023).
39. Jin, F. et al. Topological prethermal strong zero modes on superconducting processors. *Nature* **645**, 626–632 (2025).
40. Roffe, J., White, D. R., Burton, S. & Campbell, E. Decoding across the quantum low-density parity-check code landscape. *Phys. Rev. Res.* **2**, 043423 (2020).
41. Panteleev, P. & Kalachev, G. Degenerate quantum LDPC codes with good finite length performance. *Quantum* **5**, 585 (2021).
42. Chen, Z. et al. Fabrication and characterization of aluminum airbridges for superconducting microwave circuits. *Appl. Phys. Lett.* **104**, 052602 (2014).
43. Song, C. et al. Generation of multicomponent atomic Schrödinger cat states of up to 20 qubits. *Science* **365**, 574–577 (2019).
44. Google Quantum AI et al. Exponential suppression of bit or phase errors with cyclic error correction. *Nature* **595**, 383–387 (2021).
45. Jeffrey, E. et al. Fast accurate state measurement with superconducting qubits. *Phys. Rev. Lett.* **112**, 190504 (2014).

46. Heinsoo, J. et al. Rapid high-fidelity multiplexed readout of superconducting qubits. *Phys. Rev. Appl.* **10**, 034040 (2018).
47. Sunada, Y. et al. Fast readout and reset of a superconducting qubit coupled to a resonator with an intrinsic Purcell filter. *Phys. Rev. Appl.* **17**, 044016 (2022).
48. Spring, P. A. et al. Fast multiplexed superconducting qubit readout with intrinsic Purcell filtering using a multiconductor transmission line. *PRX Quantum* **6**, 020345 (2025).
49. Sank, D. et al. System characterization of dispersive readout in superconducting qubits. *Phys. Rev. Appl.* **23**, 024055 (2025).
50. McEwen, M. et al. Removing leakage-induced correlated errors in superconducting quantum error correction. *Nat. Commun.* **12**, 1761 (2021).
51. Miao, K. C. et al. Overcoming leakage in quantum error correction. *Nat. Phys.* **19**, 1780–1786 (2023).

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Methods

Code construction

We construct the [[18, 4, 4]] BB code following ref. 34. We define two binary matrices: $x = S_l \otimes I_m$ and $y = I_l \otimes S_m$, where I_m is an identity matrix of size $m \times m$ and S_l is a cyclic shift matrix of size $l \times l$, with the ith row having a single non-zero entry in the column $(i+1 \bmod l)$. Using these matrices, we define two polynomials: $A(x,y) = x^{a_1} + y^{a_2} + y^{a_3}$ and $B(x,y) = y^{b_1} + x^{b_2} + x^{b_3}$. The X-type and Z-type check matrices are then given by $H_X = [A|B]$ and $H_Z = [B^\top|A^\top]$, where '|' denotes horizontal matrix concatenation and '$\top$' denotes matrix transposition. The matrix H_X is composed of the left part (A) and the right part (B), which determine the connections between X-type check qubits and the L-type and R-type data qubits, respectively. Similarly, the connections between Z-type check qubits and the L-type and R-type data qubits are determined by $B^\top$ and $A^\top$.

To construct the [[18, 4, 4]] BB code in our experiment, we select $l = 3$ and $m = 3$, with $[a_1, a_2, a_3] = [1, 0, 2]$ and $[b_1, b_2, b_3] = [1, 0, 2]$. On the basis of these parameters, we derive the corresponding check matrices and compute the code parameters $k = 4$ and $d = 4$. Among the 18 check operators, only 14 are independent. To reduce the experimental complexity, we remove four redundant check operators measured by check qubits X^2, X^8, Z^3 and Z^4, and retain only the independent ones. Since the 14 stabilizers are independent, removing one additional stabilizer introduces an extra logical qubit, at the potential cost of reducing the code distance. To construct the [[18, 6, 3]] punctured BB code, we further remove the stabilizers measured by check qubits X^5 and Z^5, thereby increasing the number of encoded logical qubits from four to six and reducing the code distance from four to three.

For the BB code, we select a set of logical Pauli-X and Pauli-Z operators for the four encoded logical qubits:

$$\begin{aligned}
\bar{X}_1 &= X(L^0, L^2, L^4, L^5, L^6, L^7)\\
\bar{Z}_1 &= Z(L^0, L^2, L^3, L^4, L^8, R^0)\\
\bar{X}_2 &= X(L^1, L^2, L^3, L^4, L^6, L^8)\\
\bar{Z}_2 &= Z(L^0, L^2, L^4, L^5, L^6, L^7)\\
\bar{X}_3 &= X(L^0, L^1, L^6, R^0)\\
\bar{Z}_3 &= Z(L^1, L^2, L^7, R^1)\\
\bar{X}_4 &= X(L^0, L^1, L^4, L^5, L^6, R^1)\\
\bar{Z}_4 &= Z(L^0, L^1, L^6, R^0).
\end{aligned}$$

Here the notation $X(L^{l_1}, L^{l_2} \ldots R^{r_1}, R^{r_2} \ldots)$ denotes the tensor product of Pauli-X operators acting on the data qubits $L^{l_1}, L^{l_2} \ldots$ and $R^{r_1}, R^{r_2} \ldots$. A similar notation is used for the logical Pauli-Z operators. The logical Pauli operators for the punctured BB code are provided in Supplementary Section I.A.

Syndrome measurement circuit

The syndrome measurement circuit used in our work adopts the design proposed in ref. 34. We note that the used syndrome measurement circuit is not strictly distance-preserving. However, as demonstrated by the numerical simulations in ref. 34, it is nearly distance preserving: the circuit-level distance d_{cir} is close to the code distance d for all the considered codes and in some instances satisfies $d_{\text{cir}} = d$ (see table 1 of ref. 34). Our calculations yield $d_{\text{cir}} = 3$ for the [[18, 4, 4]] BB code. Since both distance-3 and distance-4 codes can correct up to $\lfloor (d-1)/2 \rfloor = 1$ error, the reduction from $d = 4$ to $d_{\text{cir}} = 3$ does not change the number of errors the [[18, 4, 4]] BB code can correct.

Data analysis

Similar to ref. 22, we convert the measurement outcomes of check qubits into stabilizer values (Supplementary Section III.A), since the check qubits are not reset between syndrome cycles. For the first cycle, the stabilizer values are directly given by the measurement outcomes of check qubits. For the final cycle, the stabilizer values are extracted from the outcomes of the data qubit readout. For the middle cycles, the measurement outcomes of the check qubits do not directly represent the stabilizer values since the check qubit states are not reset to $|0\rangle$ after executing the preceding cycle. Instead, the stabilizer values in cycle t are determined by the parity difference between the measurement outcomes of check qubits in cycles t and $t-1$. Thus, if consecutive cycles yield the same outcomes, the stabilizer value is 0; otherwise, the stabilizer value is 1.

Extracting logical error per cycle

With the obtained logical error probabilities $P_{\text{L}}(t)$ over varying cycles t, we fit data points $\{(t, P_L(t))\}_{t=1}^{6}$ using the functional form

$$P_L(t) = 1 - A(1 - p_L)^t \tag{1}$$

to extract the logical error p_{L} per cycle. To linearize this model, we define $a = \log[A]$ and $b = \log[1 - p_L]$, transforming the model into

$$y(t) = \log[1 - P_L(t)] = a + b \times t. \tag{2}$$

We perform a linear fitting using the least-squares method to determine the best-fit parameters a^* and b^*, as well as the standard error $\sigma(b)$. Finally, we exponentiate the fitted parameters to recover the original model parameters $A = \mathrm{e}^{a^*}$ and $p_L = 1 - \mathrm{e}^{b^*}$. To estimate the error bar for p_{L}, we apply error propagation to b^*, yielding

$$\sigma(p_L) = \left|\frac{\mathrm{d}p_L}{\mathrm{d}b}\right| \sigma(b) = e^b \sigma(b) = (1 - p_L)\sigma(b). \tag{3}$$

Numerical results

We simulate the logical error performance of the BB code and punctured BB code as functions of the number of cycles $t = 1 \ldots 6$. After decoding the error syndromes, we compute the logical error probabilities $P_{\text{L}}(t)$ versus t, and then fit the data to extract the simulated logical error per cycle. The fitting for the BB code (Fig. 3a,b) yields a logical error per logical qubit per cycle of $(8.24 \pm 0.18)\%$ and $(8.91 \pm 0.37)\%$ for logical Z bases and X bases, respectively. For the punctured BB code, we obtain a logical error per logical qubit per cycle of $(7.16 \pm 0.07)\%$ (logical Z bases) and $(8.18 \pm 0.14)\%$ (logical X bases) (Extended Data Fig. 3).

Physical qubit error estimation

Following ref. 22, we estimate the accumulated physical-qubit error rates as functions of the number of executed cycles m. For the physical-qubit Z-basis state preservation, the accumulated error rate $\varepsilon_{\text{p}}^{\text{z}}(m)$ for a Z-basis state over m cycles is estimated by

$$\varepsilon_{\text{p}}^{\text{z}}(m) = [1 - \exp(-t_{\text{c}} \times m/T_1)]/2, \tag{4}$$

where the cycle duration $t_{\text{c}} = 1.895\ \mu\text{s}$ and the largest measured relaxation time $T_1 = 57.8\ \mu\text{s}$. For the physical-qubit X-basis state preservation, the accumulated error rate $\varepsilon_p^{\text{x}}(m)$ is

$$\varepsilon_p^{\text{x}}(m) = [1 - \exp(-t_{\text{c}} \times m/T_2)]/2, \tag{5}$$

where T_2 is estimated via $1/T_2 = 1/(2T_1) + 1/T_\phi$, and the largest value $T_2 = 46.04\ \mu\text{s}$. The physical error rates per qubit per cycle are, thus, $\varepsilon_p^{\text{z}}(m = 1) = 1.61\%$ and $\varepsilon_p^{\text{x}}(m = 1) = 2.02\%$ (Fig. 3). To compare with the BB code that encodes four logical qubits, we plot the accumulated errors for four independent physical qubits, estimated as

$$P_p^i(m) = 1 - (1 - \varepsilon_p^i(m))^4,\ i \in \{\text{x}, \text{z}\}, \tag{6}$$

which are shown in Fig. 3a,b (dash–dotted lines).

Error budget

To quantify the relative contribution of each physical error type to the detection and logical errors, we conduct an error budget analysis following the approach in refs. 29,52. For each error type, we remove it from the noise model and recalculate the detection and logical error rates; the reduction relative to the full model (with all error sources included) is taken as a measure of its contribution. We normalize these reduction values across all error types to yield the relative contributions.

Logical performance projection

To estimate the logical performance of BB codes with distances ranging from $d=4$ to $d=12$ (Extended Data Table 1), we simulate each code under varying error suppression factors s. For each s and d, we simulate $t=6$ cycles of syndrome measurements, decode the error syndromes and compute the logical error probability $P_L(6)$. We then estimate the logical error rate per cycle as $p_L = 1-(1-P_L(6))^{1/6}$.

Decoding

We decode the error syndromes using the BP-OSD decoder[40,41] extended to the circuit-based depolarizing noise model (Supplementary Section I.C). We first simulate each scenario involving exactly one faulty operation in the syndrome measurement circuit and record the resulting error syndromes. Next, we group together faulty operations that yield the same error syndrome and compute the sum of their associated error probabilities (Supplementary Table 1). After this preparation stage, the BP-OSD decoder is applied to the error syndromes to infer the most likely error configuration by seeking a solution with the highest reliability in the formulated optimization problem. For decoding, we configure the BP-OSD decoder with a maximum of 10,000 iterations and a scaling factor, using the combination sweep method with a search depth of 7.

Data availability

The data presented in the figures and that support the other findings of this study are publicly available via Zenodo at https://doi.org/10.5281/zenodo.17706106 (ref. 53). Source data are provided with this paper.

Code availability

The data analysis and numerical simulation codes for this study are publicly available via Zenodo at https://doi.org/10.5281/zenodo.17706106 (ref. 53).

References

52. Bluvstein, D. et al. A fault-tolerant neutral-atom architecture for universal quantum computation. *Nature* **649**, 39–46 (2026).
53. Wang, K. et al. Demonstration of low-overhead quantum error correction codes. *Zenodo* https://doi.org/10.5281/zenodo.17706106 (2025).

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Author contributions

K.W. and C.Z. carried out the experiments under the supervision of C.S. and H. Wang. J.C. and Y. Wang designed the device, and J.C. and H.L. fabricated the device under the supervision of H. Wang. Z.W. designed the control and measurement electronics. Z.L. designed the error correction codes and performed the numerical simulations under the supervision of D.-L.D. Z.L., Z.-Z.S., W.L., Q.Y., S.J., Y.M., P.-X.S. and D.-L.D. conducted the theoretical analysis. All authors contributed to the experimental setup, the discussions of the results and the writing of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41567-025-03157-4.

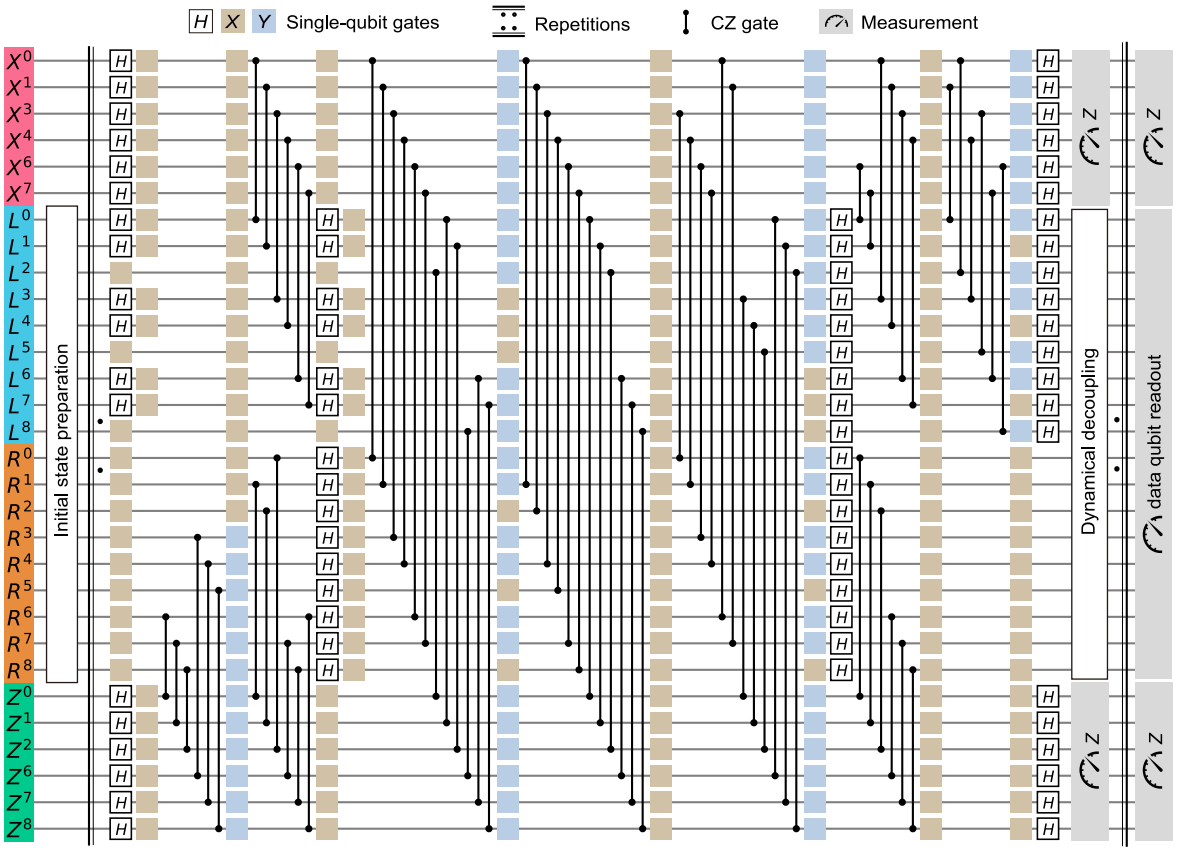
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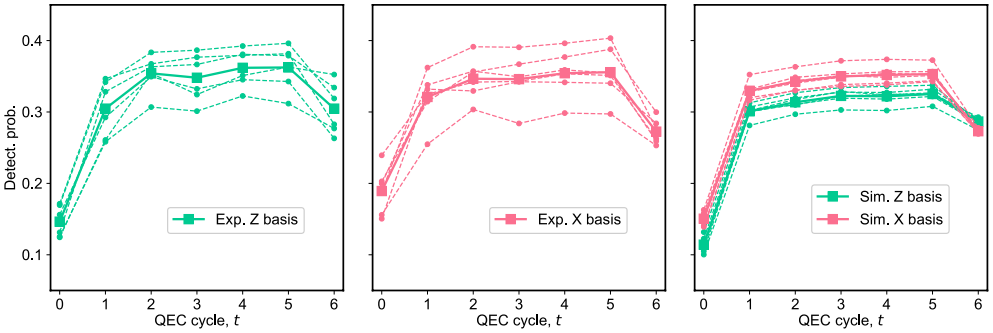
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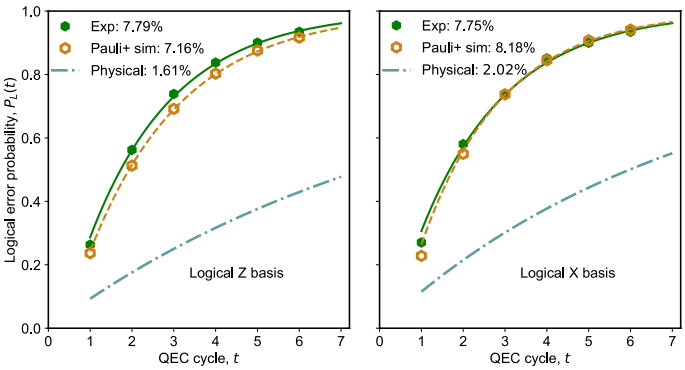
Extended Data Fig. 1 | Quantum circuit used to perform repeated stabilizer measurements for the [[18,6,3]] punctured BB code. A full syndrome cycle comprises seven layers of CZ gates, interleaved with single-qubit operations, to achieve the simultaneous extraction of all *X*- and *Z*-type stabilizers.

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Extended Data Fig. 2 | Detection probability for stabilizers over seven cycles for the [[18,6,3]] punctured BB code. Each data point is obtained from over 40,000 experimental instances. The dotted lines indicate the detection probability for each individual stabilizer, and the solid line shows the average detection probability across all stabilizers of the *Z*-type or *X*-type.

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Extended Data Fig. 3 | Accumulated logical error probabilities as functions of the number of cycles for the [[18,6,3]] punctured BB code. Each data point represents over 40,000 experimental instances after leakage rejection.

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Extended Data Table 1 | Parameters of bivariate bicycle codes used for logical performance projection

[[n, k, d]]	Encoding rate	ℓ, m	A	B
[[18, 4, 4]]	1/ 9	3, 3	$x + y^0 + y^2$	$y + x^0 + x^2$
[[36, 4, 6]]	1/ 18	6, 3	$x^1 + y^0 + y^1$	$x^2 + y^2 + y^3$
[[54, 4, 8]]	1/27	9, 3	$x^1 + y^0 + y^2$	$y^1 + x^6 + x^5$
[[90, 8, 10]]	1/ 22.5	3, 15	$x^0 + y^1 + y^5$	$y^3 + x^1 + x^2$
[[144, 12, 12]]	1/ 24	12, 6	$x^3 + y^1 + y^2$	$y^3 + x + x^2$

A code with parameters [[*n*, *k*, *d*]] encodes *k* distance-*d* logical qubits into *n* data qubits. In this construction, a total of 2*n* qubits are used, including *n* data qubits and *n* check qubits. The encoding rate is evaluated as *k*/2*n*.